MIL-C-62135A(AT)
21 May 1976
SUPERSEDING
ML-C-62135(AT)
27 October 1970

MILITARY SPECIFICATION

CRANE: MATERIAL HANDLING

This specification is approved for use by all Department and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a revolving type crane having a lifting call capacity of not less than 2,500 pounds at its maximum reach.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

RR-W-410	-	Wire Rope and Strand.

Military

MIL-V-173	-	Varnish, Moisture-and-Fungus-Resistant (For the Treatment of Communications, Electronic, and Associated Electrical Equipment).
MIL-P-514	-	Plates; Identification, Transportation-Data and Blank.
MIL-W-83420	-	Wire Rope, Flexible for Aircraft Control.

Beneficial comments (recommendations additions, deletions) and any pertinent data which may be use in improving this document should be addressed to: U.S. Army Tank-Automotive Materiel Readiness Command by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC-3950

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STANDARDS

Military

MIL-STD-105	- Sampling Procedures end Tables for Inspection and Attributes
MIL-STD-193	- Painting Procedures, Tactical Vehicles (Tracked and Wheeled).
MIL-STD-461	- Electromagnetic Interference Characteristic Requirements for Equipment
MIL-STD-721	- Definitions of Effectiveness Terms for Reliability, Maintainability, Human Factors for Safety.

DRAWING

7722347	- Receptable.
11651265	- Crane Assembly.
11651259	- Support.
11651250	- Materials Handling Kit.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Initial production crane. The initial production crane and all cranes produced thereafter shall meet the performance requirements of this specification. The first crane produced from production tooling shall be fully representative of cranes supplied under this contract. This crane shall be submitted to the Government for inspection and tests. Production and acceptance of production cranes shall be withheld until such time that the initial production crane has successfully demonstrated the performance specified herein.

3.1.1 Reliability (minimum). The reliability of the material handling crane shall have a specified value (SV) of 50 mean cycles between mission failure (MCBMF) to be demonstrated in accordance with the mission profile (see 6.4) during test. For calculation of MCBMF, a failure shall be defined as a condition which:

a. Prevents operations.
b. Reduces performance below essential levels.
c. Indicates to the crew that further operation would be unsafe.
d. Indicates to the crew that further operations might result in extensive damage to the equipment.

NOTE: Any of the above four conditions which cannot be corrected by the crew in 30 minutes using the tools and parts normally carried on the M877 Cargo Vehicle constitutes a failure. Maintenance and human induced failures are excluded.

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3.1.2 Durability (minimum). When installed on the M520 cargo vehicle the crane assembly shall have a .90 probability of not requiring removal from the vehicle for repair during 125 hours of crane operation (125 crane missions, see 6.4) without requiring replacement of motors or crane brake.

3.1.3 Maintainability. When maintenance procedures shown in the technical manuals are followed and the crane is subject to endurance testing, or operation on the vehicle as defined in the mission statements the demonstrated maintainability shall be as follows:

(a) Means Operator Preventive Maintenance Time shall not exceed .50 manhours per mission. This time shall not be included in scheduled maintenance time.
(b) Maximum Operator Corrective Maintenance Time shall not exceed .15 manhours per mission without being classified as a mission failure.
(c) The ratio of total corrective and scheduled maintenance manhours to operating hours shall not exceed .12.
(d) The ratio of total scheduled maintenance manhours to total operation hours shall not exceed .10.
(e) The ratio of total corrective maintenance manhours to operating hours shall not exceed .02.
(f) Mean manhours to perform a corrective maintenance action (MTTR) shall not exceed 1.00.
(g) The specified mean cycles between maintenance actions (MCBMA) shall not be less than 21 cycles.

3.1.3.1 Maintainability Failure. A maintainability failure is defined as any unscheduled adjustments, repair or replacement. To calculate MCBMA, consult definitions in MIL-STD-721.

3.1.4 Interchangeable components. The electric motor assemblies provided for turntable, cable hoist, and boom lift shall be identical.

3.2 Materials. Materials specified herein in referenced drawings, specifications, and standards shall be free from all defects and imperfections that might affect the serviceability and function of the finished product.

3.3 Crane assembly. The crane assembly shall be in accordance with 11651265. The assembly shall include winches, wire rope, load block, lifting hook, sheaves, controls, wiring, and all items necessary for its complete adaptation. All direct load carrying components of the crane shall have a safety factor of not less than two, based on the yield point of the material used. The liftoff weight of the complete crane assembly shall not exceed 875 pounds. The crane liftoff configuration shall include features of removal from the truck as a separate unit, bolts securing the crane to the vehicle, with waterproof electrical connections between the truck and crane.

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3.3.1 Boom, line, and block. With the boom parallel to the ground, the boom length measured from the centerline of rotation to the center of the load hook, shall be not less than eleven feet, and the overall length not more than eleven and one half feet. In the event of an electrical failure means shall be incorporated to manually return the boom to the traveling position. The boom assembly shall be adaptable to 11651259 to permit securing the boom during vehicle travel operations. The wire ropes provided shall be corrosion resistant (tin or zinc coated) cable conforming to type I, composition A, construction 7 x 19, of MIL-W-83420. Hoist cable to be 1/4 inch dia. Boom cable to be 5/16 dia.

3.3.2 Location of crane. The crane assembly shall be mounted on the left hard rear wheel well axle forwarded from the centerline of the rear axle of the M520 cargo vehicle.

3.3.3 Swing column. The height and width shall be not more than 45 inches and 23 inches respectively.

3.3.4 Seals. All joints and openings requiring sealing shall be sealed against water and contaminant entry, as well as lubricant leakage during operation storage. When installed, the crane shall withstand all conditions of vehicle operations which includes 60% grade and 30% side slopes, and there shall be no evidence of lubricant leakage.

3.3.5 Lubricants. Lubricants, type and grade to be used, shall be as specified in vehicle lubrication chart.

3.3.6 Crane base. The crane base mounting dimensions shall be in accordance with 11651265. The swing column shall be bearing mounted to the crane base. The bearings shall be of a size and type to withstand the axial loads exerted on the column under static and dynamic conditions, while lifting the rated payloads.

3.3.7 Operator controls. The controls shall be dead man type or spring centered, normally opened electrical switches, enclosed within a waterproof pendant assembly, and having 20 feet of cable from the crane base. The cable when not in use shall be stowed on top of boom.

3.3.8 Power source. The crane shall be powered by the vehicle's 24 volt direct current electrical system through an electrical supply cable attached to the center of the crane's base. The cable length from the crane base shall be not less than 15.00 inches nor more than 18.50 inches with plug compatible with 7722347. The complete electrical system which includes motors, brake solenoids, cables, pendant assembly, and relay controls to provide a complete installation, shall be furnished.

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3.3.8.1 Fungus resistance. Electrical connections, including terminal and circuits, shall be coated with varnish conforming to MIL-V-173 except that:

(a) Components and elements inherently inert to fungi or hermetically sealed need not be treated.
(b) Components and elements whose operation shall be adversely affected by the application of varnish shall not be treated.

3.4 Performance. All electrical equipment shall be actuated through the controls mounted within the remote pendant. The crane assembly shall demonstrate smooth and quiet operation, ease end flexibility of operation, and versatility of performance. The entire system shall meet all requirements specified herein in temperatures from minus 40° to plus 125° F. Any two operations of the crane shall be capable of being performed simultaneously.

3.4.1 Radio Interference Suppression. The crane, when operated in its intended application as a system, shall meet the requirements for the tactical and special purpose vehicle provisions of MIL-STD-461, notice 4.

3.4.2 Rotation. The crane shall travel through 360 degrees clockwise, and counter-clockwise, at a speed of 1 revolution per minute, ±10 seconds, on level ground. The turntable shall rotate the crane assembly smoothly with rated payload and the vehicle positioned on a compound slope, longitudinal and transverse, of 10 percent. The turntable drive shall be equipped with a self-energized brake capable of retaining the turntable in a fixed position. The self energized brake shall retain a 2500 pound load under the following conditions:

(a) Boom at maximum extension.
(b) Vehicle positioned on 10% grade.
(c) Boom positioned at 90 degrees to 10% grade.
(d) With drive motor de-energized, a means shall be provided to override the brakes and manually rotate the turntable in the event of an electrical failure.

3.4.3 Elevation and hoisting. The boom assembly shall elevate through a minimum angle of 0 degrees to 80 degrees. The load hoisting cable shall demonstrate a hoist hook speed of 9 to 12 feet per minute when hoisting a 2500 pound load. The boom and hoist drive shall be equipped with self energizing brakes, shall retain a 2500 pound load in a fixed position when the drive motors are de-energized. A device shall be provided to override the brakes to permit manual lowering of the load and boom in the event of an electrical failure. The emergency handcrank shall be stowed on top of the electrical switching components enclosure.

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3.5 Safety features All exposed parts of the crane which are subject to extremes in temperature or which are energized electrically, or are so located as to be a hazard to operating personnel, shall be insulated, inclosed, and guarded. A load sensing device shall be provided to preclude lifting of loads in excess of the rated capacity of the crane. The crane shall be capable of lifting 110% of rated load (2,750 lbs.) at the maximum and minimum load radius of the crane. The load radius is the distance between the center of rotation and the center of the load block, measured on a horizontal plane with the load suspended. A device shall be provided to override the brakes to permit manual lowering of the load and boom in the event of an electrical failure.

3.6 Data plates. All data plates shall conform to Type III, Class 2, of MIL-P-514.

3.6.1 Operational data plate A data plate shall be mounted on the crane swing column instructing the operator of the correct procedure to follow in the event there is an electrical failure. All operational and caution data plates shall be mounted in locations readily visible to the crane operator. Plates shall contain all necessary warning type information to assure proper operation including the crane's rated lifting limitations. The operational data plate for operating the crane in case of a power failure shall include the "model number", "part number 11651265," "serial number" (to be stamped in), "capacity 2,500 pounds" at any boom angle, as well as any appropriate instructions.

3.7 Treatment and painting. Treatment and painting shall be in accordance with MIL-STD-193.

3.8 Workmanship. The assembly shall be constructed and finished thorough in a workmanlike manner, particular attention shall be paid to throughness of painting, plating, bolted on assemblages, welding and brazing. Electric circuits shall be located in a manner so as to prevent possible damage by rubbing on adjacent lines or appendages. All parts and assemblages shall be free from burrs, sharp edges or any condition which may present a safety hazard to operating and maintenance personnel.

3.9 Installation instruction and operators' manual. The contractor's commercial operator's manual with a breakdown of his commercial part numbers and installation instructions shall be furnished. Each kit shall be packaged in a waterproof envelope with additional copies furnished as follows for each contract:

U.S. Army Tank-Automotive Command
Warren, MI 48090

ATTN: AMDTA-RE (2 copies)
ATTN: DRSTA-QK (2 copies)
ATTN: DRSTA-M (2 copies)

NOTE: The contract number shall be provided on the cover of the manual.

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4. Quality Assurance Provisions

4.1 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Contractor's Inspection or Quality Control System. The contractor shall provide and maintain an effective inspection or quality assurance system acceptable to the Government covering the supplies under the contract. A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The contractor shall obtain from the Government approval of any subsequent change to the submitted procedures prior to implementing the change.

4.1.2 Government Verification. All quality assurance operations performed by the contractor shall be subject to Government verification at unscheduled intervals. Verification shall consist of surveillance of (a) the operations to determine that practices, methods, and procedures of the written inspection plan are being properly applied, and (b) Government inspection to measure quality of product offered for acceptance. Deviation from the prescribed or agreed upon procedures, or instances of practices which might have an affect upon the quality of the product, will be immediately called to the attention of the contractor. Failure of the contractor to promptly correct defects discovered shall be cause for suspension of acceptance until corrective action has been taken, or until conformance of product to specified criteria has been demonstrated.

4.2 First Production Crane Inspection. During fabrication of the first production crane, an in-process inspection will be conducted by Government representative(s) to evaluate conformance of materials and workmanship to requirements of specified documents. These inspections shall be made at the contractor's facility prior to the application of primer and paint. Processing and welding procedures, quality control system and inspection records will be reviewed and evaluated during this inspection.

4.2.1 Completed First Production Crane Inspection. The first production crane, when complete, shall be inspected by a Government representative at the contractor's plant subsequent to contractor's inspection to determine conformance to contract and specification requirements. Contractor shall further provide any other inspection assistance as may be required. At the time of this inspection the contractor shall make available his inspection plan, inspection records, and certifications pertinent to the crane and its components.

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4.2.1.1 Repair of Defects. Defects found as a result of above inspection shall be corrected by the contractor at no cost to the Government. Failure of the contractor to promptly correct defects shall be cause for suspension of acceptance of cranes until corrective action has been accomplished.

4.2.3 Crane Disposition. Upon completion of inspection and acceptance of the first production crane, the crane shall remain at the manufacturing facility as a production sample. The contractor shall service and maintain the crane during this period while in storage. At the discretion of the contracting officer the crane may be released for shipment.

4.2.4 Final Approval and Acceptance. Final approval and acceptance by the Government of the first production crane shall be withheld until the initial production test has been completed and final determination has been made regarding conformity of the crane to contractual requirements, including but not limited to, workmanship and materials.

4.3 Initial Production Test. Unless otherwise specified (see 6.2) the crane(s) shall be selected by the Government from the first months production and shall be installed on an M520 Cargo vehicle. Cranes shall be subjected to examinations and tests listed in Tables I and II. Test will include 200 miles of paved road, 400 miles of secondary roads and 400 miles of cross country operation (1000 miles of roadtesting) at a Government approved test site. A total of 125 crane missions shall be performed in the following manner:

a. 25 Crane missions using paved road
b. 50 Crane missions using secondary road
c. 20 Crane missions using level cross country
d. 5 Crane missions using level cross country mud
e. 25 Crane missions using hilly cross country

The contractor shall support the test by providing any crane parts necessary for completion of such tests. Delay in test completion caused by crane breakdown or failure of the contractor to adequately support the crane with parts during the tests shall not be the basts for adjustment of contract delivery schedule or contract price.

4.3.1 Reliability Verification. To determine conformance to the reliability specified in para. 3.1.1a Minimum Acceptable Value (MAV) of 35 MCBMF will be demonstrated during test, with a confidence level at or above 50 percent.

4.3.2 Durability Verification. To determine conformance to paragraph 3.1.2, a loading and unloading cycle consisting of sequences of six 2,500 pound loads shall be performed in accordance with instructions shown on Drawing 11651269, Plate Instructions. Complete four load and unload cycles of the six 2,500 pound loads utilizing a minimum of 45 minutes for each cycle to prevent overheating of the motors. After each four load and unload cycles of the six 2,500 pound loads, the motors shall be allowed to cool down to a temperature safely touchable by hand (approximately 45 minutes cool down time). To verify paragraph 3.1.2., at least 50% of the test sample must meet the requirements. This verification will be demonstrated during test using a minimum of 125 cycles.

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4.3.3 Maintainability Verification. Requirements of para. 3.1.3 shall be verified during applicable tests. Requirement of para. 3.1.3(g) shall be verified by demonstrating a minimum acceptable value (MAV) of 14 MCBMA with a confidence level of at least 50%.

4.3.4 Test Failure. Failure of the test crane to meet minimum performance requirements, failure to comply with the requirements of a workmanship or materials nature and failure to manufacture the crane in accordance with the technical data revealed during or as a result of this test, shall be cause for rejection of the crane. Further, the Government may refuse to continue acceptance of subsequent production cranes until evidence has been provided by the contractor that corrective action has been taken to eliminate the defects. Any defects found during or as a result of this test shall be prima facie evidence that all cranes already accepted prior to completion of test are similarly defective unless evidence satisfactory to the Government is furnished by the contractor that they are not similarly deficient. Such defects on all cranes shall be corrected by the contractor at no cost to the Government.

4.4 Inspection Provisions.

4.4.1 Lot Size. An inspection lot shall consist of all cranes from one weeks production, submitted at one time for acceptance inspection.

4.4.1.1 Sampling for Inspection. For the purpose of visual, dimensional and primary function inspection, a representative sample shall be selected from each inspection lot in accordance with inspection level II of MIL-STD-105. Before sampling may be accomplished by the contractor, the contractors shall 100% inspect the first ten cranes to establish a process average (see 6.7) to allow normal sampling in accordance with MIL-STD-105.

4.4.1.2 Examinations. Visual, dimensional, and primary functional examination shall consist of examination of the crane for conformance to the applicable drawings, contract, and this specification. Examination shall be performed against the classification of defects and with the acceptable quality levels shown in Table I. The following constitutes a part of the classification of defects (see 6.3 and 6.8):

(a) Any drip constitutes a major defect when the crane has been standing idle and the components are at ambient temperatures.

(b) Any droplet that occurs at a static fit metal to metal or gasket combination shall constitute a major defect.

4.4.1.3 Unclassified Defects. All defects having no bearing on function, safety, interchangeability or life, but are considered departures from good workmanship will be noted in writing. Workmanship deficiencies falling within this category and recurring in five consecutive lots or ten lots or more within a thirty-day period, will be added to the minor classification with no increase to AQLs.

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4.4.1.4 Recurring Major Deficiencies. A major deficiency (see 6.8) is recurring when the same defect occurs more than once in the same sample or when the defect occurs in two successive samples. A major defect may be considered recurring when the historical inspection records reflect such a condition. Recurring major deficiencies shall be cause for the entire lot or lots to be inspected for the recurring deficiencies. The deficiencies shall be corrected when found.

4.4.1.5 Recurring Minor Deficiencies. A minor deficiency (see 6.8) is recurring when it occurs more than twice in the sample or when the defect occurs in four successive samples. Recurring minor deficiencies shall be cause for the entire lot or lots to be inspected for recurring deficiency. Correction of all deficiencies shall be accomplished prior to acceptance.

TABLE I - CLASSIFICATION OF QUALITY DEFECTS

Defect Number	Defects	Method of Inspection
Major	AQL 1.0% Defective	
101	Electric Motors: Improper installation, Misalignment	Visual and Functional
102	Control Switches: Malfunction	Visual and Functional
103	Electric Wiring: Improperly Installed, Damaged	Visual and Functional
104	Welding: Undercut, Pinholes, Inadequate	Visual
105	Dimensions: Mounting and Affecting Interchangeability	Visual and Gage
Minor	AQL 2.5% Defective	
201	Workmanship: Adjustment, Installation of Components	Visual and Functional
202	Instructions: Improper, Not Legible	Visual
203	Paint: Improper Application., Color	Visual

4.4.2 Examination Failure. If a crane is found to contain any defects, it shall be rejected. Rejected cranes may be repaired and re-examined for compliance to requirements.

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4.5 Classification of Tests.

(a) Acceptance Test: 4.5.1 through 4.5.1.9

(b) Performance Test: 4.5.2 through 4.5.2.3

(c) Comparison Test: 4.5.3 and 4.5.3.1

4.5.1 Acceptance Tests. Each crane assembly shall be subjected to the acceptance tests listed in Table II.

4.5.1.1 Materials and Component Check. To determine conformance to 3.1.4 and 3.2 all materials shall be examined for damage.

4.5.1.2 Crane Assembly Check. To determine conformance to 3.3, the crane assembly shall be examined for completeness of assembly and weight requirements.

4.5.1.3 Boom, Line and Block Check. To determine conformance to 3.3.1, each boom, line and block shall be checked for meeting requirements.

4.5.1.4 Swing Column Check. To determine conformance to 3.3.3, the column shall be checked to height and width requirements.

4.5.1.5 Seals Check. To determine conformance to 3.3.4, each assembly shall be checked for leaks greater than a weep or seep (see 6.3).

4.5.1.6 Lubricants Check. To determine conformance to 3.3.5, the lubricants used shall be checked for proper tape and level.

4.5.1.7 Crane Base Check. To determine conformance to 3.3.6, each crane base shall be checked for proper mounting, bearing size and type.

4.5.1.8 Operator's Control Check. To determine conformance to 3.3.7, the operators controls shall be checked for proper installation, and cable checked for length.

4.5.1.9 Power Source and Fungus Resistance Check. To determine conformance to 3.3.8 and 3.3.8.1, the power source shall be checked for completeness of assembly and installation.. Prior to installation, the electrical components shall be checked for meeting fungus resistance requirements.

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TABLE II - TEST SCHEDULE

	Requirement	Inspection	IPT	Acceptance	Comparison
Reliability	3.1.1	4.3.1	X		X
Durability	3.1.2	4.3.2	X		
Maintainability	3.1.3 & 3.1.3.1	4.3.3	X		X
Materials & Components	3.2 & 3.1.4	4.5.1.1	X	X	X
Crane Assembly	3.3	4.5.1.2	X	X	X
Boom Line and Block	3.3.1	4.5.1.3	X	X	X
Swing Column	3.3.3	4.5.1.4	X	X	X
Seals	3.3.4	4.5.1.5	X	X	X
Lubricants	3.3.5	4.5.1.6	X	X	X
Crane Base	3.3.6	4.5.1.7	X	X	X
Operations Control	3.3.7	4.5.1.8	X	X	X
Power Source & Fungus Resistance	3.3.8 & 3.3.8.1	4.5.1.9	X	X	X
Performance Tests	3.4	4.5.2	X	X	X
Radio Interference	3.4.1	4.5.2.1	X		X
Rotation	3.4.2	4.5.2.2	X	X	X
Elevation & Hoisting	3.4.3	4.5.2.3	X	X	X
Comparison		4.5.3			X
Safety Features	3.5	4.6	X	X	X
Data Plates	3.6	4.7	X	X	X
Operation Data Plates	3.6.1	4.7.1	X	X	X
Treatment and Painting	3.7	4.8	X	X	X
Packaging and Packing	5	4.9		X	

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4.5.1.10 Test Failure. If a crane fails to pass any acceptance test specified herein, the Government inspector shall stop acceptance of subsequent cranes until evidence has been provided by the contractor that corrective actions have been accomplished.

4.5.2 Performance Test. To determine conformance to 3.4, each complete crane assembly shall have all electrical equipment actuated and checked for functional requirements.

4.5.2.1 Radio Interference Suppression Test. To determine conformance to 3.4.1, the crane assembly shall be checked for meeting radio interference requirements.

4.5.2.2 Rotation Test. To determine conformance to 3.4.2, the crane assembly shall be operated in both directions and checked for meeting performance requirements. During the operation of the turntable, the self energized brake performance shall be tested using an actual or simulated load, actual or simulated positions and checked for meeting performance requirements.

4.5.2.3 Elevation and Hoisting Test. To determine conformance to 3.4.3, the boom assembly shall be operated through the degrees of elevation, and cable speed checked for meeting requirements. The boom and cable with specified load shall retain the load in a fired position when the self energized brake is actuated. The override device shall permit manually lowering the load and boom in the event of an electrical failure.

4.5.3 Comparison Tests. The Government may select crane assembly any time during a production contract period and subject crane assembly to all tests listed on Table I and Table II to reveal defects of a workmanship or materials nature that may reduce the effective operation of these items in the field and compare existing quality with previous standards. These tests shall be conducted at Government laboratories or proving grounds designated by the contracting officer. The crane assembly mounted on the M520 vehicle shall meet all requirements of this specification.

4.5.3.1 Test Failure. Failure of any crane assembly tested to comply with any of the requirements specified in the contract, or any, major defect of a workmanship or materials nature occuring during or as a result of the test cycle, shall be considered cause for refusal to continue acceptance of crane assembly by the Government until objective evidence has been shown that corrective action has been taken to eliminate the condition which caused the failure. Any defects found during or as a result of the test shall be prima facie evidence that cranes accepted subsequently are similarly defective unless evidence satisfactory to the contracting officer is furnished by the contractor that they are not similar defects. Such defects on all crane assemblies shall be corrected by the contractor at no cost to the Government.

4.6 Safety Features Check. To determine conformance to 3.5 during crane operation, all safety features including sensing device shall be checked for meeting requirements.

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4.7 Data Plates Check. To determine conformance to 3.6, data plates shall be checked for proper type and legibility.

4.7.1 Operational Data Plate Check. To determine conformance to 3.6.1, the plates shall be checked for location instruction and warning type information. Operational test shall be conducted in accordance with instruction on plates.

4.8 Treatment and Painting Check. To determine conformance to 3.7, procedures, treatment and painting will be checked for conformance to MIL-STD-193.

4.9 Packaging and Packing Tests. To determine conformance to Section 5, the package and the pack shall be subjected to the tests specified for the prescribed packaging method, and for the prescribed packing in the referenced packing specifications.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. Cleaning, packaging, packing and marking shall be in accordance with the specified packaging data sheet or sheets (see 6.2).

6. NOTES

6.1 Intended use. The crane assemblies covered by this specification are intended for use on the M520 cargo vehicle.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Quantity of initial production test cranes assemblies required (see 4.3).
(c) Applicable packaging data sheet or sheets (see 5.1).

6.3 The following definitions shall be used for classification of defects for leaks:

(a) Weep: Any nonrecurring evidence of fluid beyond the seal.
(b) Seep: Any recurring evidence of fluid beyond the seal that does not result in formation of a droplet.
(c) Droplet: Any recurring evidence of fluid beyond the seal that results in a formation of a non-falling droplet.
(d) Drip: Any recurring evidence of fluid beyond the seal where droplet forms and fall.

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6.4 Mission Statement. Add the following paragraphs from MIL SPEC MIL-T-62132A (AT); Truck, Cargo: 8 Ton, 4x4, M520 for reference purposes: "6.5 Mission Statement. One mission shall consist of 100 miles of operation under these conditions of load, speed, terrain, and environment with special functions described."

TERRAIN	DISTANCE	OPERATING SCHEDULE RANGE MPH
Hard Surface	20 Miles	0-30
Secondary	40 Miles	0-30
Level Cross Country	16 Miles	0-30
Level Cross Country Mud*	4 Miles	0-10
Hilly Cross Country*	20 Miles	0-15
Swimming	Traverse Water Barrier	5 Minute Operation

60% Grades: 1/10 of 1% Hilly Cross Country Operation w/o Towed Load.
30% Side Slopes: 1/10 of 1% Hilly Cross Country Operation w/o Towed Load.

Climatic Extremes
Temperature: -25 F. To 125 F. Ambient
Altitude: 500' to 5,000'

Tractor Functions:

Idle: 30 Minutes
Starts: One cold, one hot at two minutes, one hot at 15 minutes after shutdown (Cold means coolant & oil at ambient, hot at maximum obtained)
*4-Wheel drive: 100% mud & swim, Cross country as required, all other operation is in 2-wheel drive.
Reverse: 10 operations at 130 feet each.

From winch (once each 10 missions) one overload attempt (100% of rated capacity followed by recovery of a load at 90% of capacity).

Cargo Carrying Function:

Operation: 50% of time at 8 ton payload, 25% of time at 4 ton payload, and 25% at no load, 10% with towed load.

When Equipped With MHE Crane: Load and unload 8 ton palletized cargo every second mission.

6.6 Typical Vehicle Life Operational Profile

Model	Renewal Life Miles - Hours	AVG M.P.H	Total Missions	Mission Miles - Hours
M520	25,000-1250	20.0	250	100-5
M559	18,000-1250	14.4	182	100-6.9
M553	12,500-1250	10.0	250	50-5

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In actual troop use, this will vary due to environmental terrain, and troop mission differences.

6.7 Process Average. Sampling may be initiated if the process average value of the first ten cranes inspected is less than the AQL specified in the Classification of Defects for major and minor defects. Process average shall be computed as follows:

$$\text{Process average} = \frac{\text{Number of Defects}}{\text{Number of cranes inspected}} \times 100$$

If the computed process average exceeds the specified AQL, 100 percent inspection shall be performed and continued until such time that the process average for ten consecutive vehicles is less than the specified AQL.

(2) Paragraph 6.8 "Definitions of Recurring Major and Minor Defects. The Government Inspector will verify that a thorough inspection of each crane is performed by the contractor not only for the listed characteristics, but also for any other departures from good workmanship. The Government Inspector will assure that all deficiencies encountered during the inspection are enumerated on the Deficiency Sheet for the crane. The defects noted on the Deficiency Sheet shall contain sufficient description to enable the Government Inspector and the contractor's representative to classify the deficiency in accordance with the classification of defects of the crane specifications and definitions contained in MIL-STD-105. Corrective action shall be taken for recurring deficiencies.

Custodian:
Army - AT

Preparing Activity
Army - AT
Project No. 3950-A196

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